Release Estimation

Reference Guide

This guide explains what the Release Estimation web app does, how to configure and use it, and how to read and act on its results.

1. PURPOSE & BACKGROUND

THE CORE QUESTION
“When will we finish this release — and how confident can we be in that date?”

That is the question this tool is built to answer. It takes your team’s velocity estimates and the size of your backlog, applies a statistical model to account for natural variation in delivery, and projects a completion date at a confidence level you choose.

The result is a structured estimate, not a forecast. Forecasting requires historical sprint data to calibrate the model. This tool works without that history — it relies instead on your team’s own assessment of their likely, worst-case, and best-case velocity. That means the quality of the output depends directly on the honesty and care of the inputs.

What the confidence level means

When the app says “80% confidence — finish by June 15”, it means: if this release were attempted many times under similar conditions, the team would finish by June 15 in 80 out of 100 of those attempts. There is a remaining 20% chance it runs over. Higher confidence pushes the date later; lower confidence brings it earlier.

What this tool does not do

- It does not track actual sprint delivery or compare estimates to actuals.
- It does not account for scope changes after the estimate is set.
- It does not replace planning — it supports it by making uncertainty explicit.
- It does not tell you whether the backlog is the right backlog.

2. SETUP & CONFIGURATION

Teams

The app supports multiple teams. Each team has its own releases, scenarios, and configuration settings. When you log in you land on the 'Your Teams' page. Click 'Open' to go to the Estimation page for a team. Once a team is open, use the sidebar to navigate between Estimation and Configuration.

Releases

A release represents one upcoming delivery you want to estimate — for example, "v2.0" or "Q3 Release". Each release contains one or more scenarios. Archive a release when it is complete to keep the list tidy. Archived releases can be restored at any time using the 'Show Archived' toggle.

Scenarios

A scenario represents one set of assumptions within a release. You might have an "Optimistic" scenario with a higher velocity estimate and a "Conservative" scenario with a lower one. The Comparison Table at the bottom of the Estimation page shows all valid scenarios side by side.

Configuration page

The Configuration page lets you set team-level defaults that pre-fill new scenarios automatically. Changing these settings does not affect scenarios that have already been created.

Unit of Work

Choose 'points' or 'issues' depending on how your team measures velocity. This label appears throughout the Estimation page.

Default Sprint Length

The sprint duration in weeks that new scenarios will start with.

Default Confidence in Projected Date

The starting confidence level slider value for new scenarios.

Default Confidence in Most Likely

The starting selection for the velocity confidence dropdown in new scenarios.

Use 'Reset to Defaults' to restore all settings to their original values.

3. ENTERING ESTIMATES

Each scenario has the following inputs. Results recalculate automatically as you change values. Click ‘Save Changes’ to store them.

Sprint Length (weeks)

How many calendar weeks your sprints run. Most teams use 1 or 2 weeks.

Total Backlog

The total amount of work remaining in the release, expressed in your unit of work (points or issues). This is the finish line.

Release Start Date

The date the first sprint of this release begins. The projected completion date is calculated forward from here.

Worst Case velocity

The lowest velocity your team could realistically deliver in a sprint. This should reflect a genuinely bad sprint — not a catastrophic one. Must be less than Most Likely.

Most Likely velocity

The velocity your team most commonly delivers. This is the central input — the peak of the bell curve.

Best Case velocity

The highest velocity your team could realistically deliver. Must be greater than Most Likely.

Extra Non-Working Days

Calendar days within the release window when the team is not working — public holidays, team events, company shutdowns. Enter the total count. These days are added to the projected timeline.

The three velocity inputs (Worst, Most Likely, Best) should reflect your team’s per-sprint throughput in a normal delivery environment. They are not targets — they are honest assessments of the realistic range.

Expert Settings (Advanced)

The ‘Expert Settings’ expander contains one additional input: ‘Standard Deviation Override’. Leave this at 0 unless you have a specific reason to override the calculated value. When set above 0, it replaces the model’s derived standard deviation with the value you enter, directly controlling the spread of the distribution.

4. THE TWO CONFIDENCE SETTINGS

There are two separate confidence-related inputs in each scenario. They control different things and are easy to confuse.

Confidence in Most Likely	Confidence in Projected Date
About your velocity estimate	About the projected completion date
How certain are you that your Most Likely velocity will actually occur?	How certain do you want to be that the team finishes by the projected date?
Controls the width of the bell curve (standard deviation)	Controls where on the bell curve the date is read from
High → tight distribution, narrow range of outcomes	High → later date, more conservative guarantee
Low → wide distribution, more uncertainty in outcomes	Low → earlier date, higher risk of running over
A property of how well your team knows its own velocity	A stakeholder decision about acceptable risk

The two settings interact. A wide distribution (Low confidence in Most Likely) combined with a high desired confidence (90%+) produces the most pessimistic projected dates — because you are saying "my velocity is uncertain AND I want a near-guarantee."

A common starting point: set Confidence in Most Likely to 'Medium' and Confidence in Projected Date to 80%. Adjust from there based on how predictable your team's velocity has been in practice.

5. READING THE RESULTS

Results appear below the inputs when all velocity estimates are valid. They update in real time as you change values.

Projected Date

The calendar date by which the team is expected to complete the release at the selected confidence level.

Confidence

The confidence level currently selected — the probability that delivery will occur on or before the Projected Date.

Sprint Completed In

The sprint number in which the work is estimated to be finished. Sprint 1 is the first sprint of the release.

Business Weeks

The number of working weeks from the start date to the projected date, excluding weekends. Use this figure — rather than the calendar date — when planning around sprints.

The narrative

Below the metrics, a plain-language sentence summarises the result: "At 80% confidence, your team will complete [Release] by [Date]. The work is estimated to complete in Sprint 8, requiring at least 18.5 points per sprint across 16 business weeks." The 'at least X per sprint' figure is the velocity at selected confidence the model is using to project the date.

Calculation Details

Expand 'Calculation Details' to see the underlying numbers: PERT weighted mean velocity, statistical standard deviation, velocity at selected confidence, raw sprints needed, and total calendar days. These are useful for checking the model's assumptions or explaining the result to others.

Validation warnings

- If Worst Case is greater than or equal to Most Likely, or Best Case is less than or equal to Most Likely, the results will not appear and a warning will prompt you to correct the values.
- If the velocity estimates are highly asymmetric (the gap between Most Likely and Worst/Best is very uneven), a warning appears suggesting you rebalance the estimates for more reliable results.
- If Confidence in Projected Date is set below 50%, a warning notes that at low confidence levels unknown factors frequently cause estimates to run over.

6. CHARTS

Expand the ‘Charts’ section within a scenario to view two charts.

Velocity Distribution (Bell Curve)

Shows the probability distribution of velocity at selected confidence based on your inputs. The curve represents the range of velocities the model considers plausible. The vertical line marks the velocity at selected confidence at your selected confidence level — the point where the shaded area to the left equals your desired confidence percentage.

Projected Date by Confidence Level

Shows how the projected completion date shifts as confidence increases from 10% to 99%. The current confidence level is marked. Use this chart to understand the trade-off between confidence and date — for example, moving from 70% to 90% confidence might push the date out by two sprints.

Scenario Comparison Chart

When two or more valid scenarios exist, a horizontal bar chart appears at the bottom of the page showing business weeks to completion for each scenario. This makes it easy to compare the impact of different assumptions at a glance.

7. TARGET DATE ANALYSIS

Expand ‘Target Date Analysis’ within a scenario to enter a deadline and get two outputs.

Confidence this date will be met

The highest confidence level at which the projected date falls on or before your target. If this is lower than your selected confidence level, the target is only achievable by accepting more risk. If the target is earlier than what any confidence level allows given current inputs, an error message appears.

Minimum velocity needed

The velocity your team would need to maintain in order to hit the target date. Shown with a delta against the current velocity at selected confidence. Green indicates your current estimate already meets the target; amber indicates a gap that needs to be closed.

Target Date Analysis is a read-only diagnostic. It does not save or change any inputs. Use it to answer “could we hit this deadline?” before committing to one.

8. SHARING & EXPORT

Expand 'Share & Export' at the bottom of the Estimation page to access three options.

Shareable Link

Generates a unique URL that anyone can open without logging in. The linked page shows a read-only view of the release with all valid scenarios and charts. Click 'Revoke Link' at any time to disable access. Each release has its own independent link.

CSV Export

Downloads a spreadsheet with one row per valid scenario, including all inputs (backlog, velocity estimates, sprint length, confidence settings) and computed outputs (projected date, business weeks, sprint number, minimum velocity). Useful for pasting into planning tools or sharing via email.

Plain-Text Summary

A formatted text block showing the key result for each valid scenario. Click the copy icon in the top-right corner of the box to copy it to the clipboard, then paste directly into a Slack message, email, or document.

9. THE MATH BEHIND THE MODEL

This section explains how the app moves from three velocity inputs to a projected date. No statistical background is required to use the app — this is for those who want to understand or explain the calculation.

Step 1 — PERT weighted mean

The three velocity inputs are combined into a single weighted average using the PERT formula:

Mean = (Worst Case + 4 × Most Likely + Best Case) ÷ 6

This gives Most Likely four times the weight of the two extremes, reflecting the assumption that the central estimate is the most informed.

Step 2 — Standard deviation

The spread of the distribution is controlled by the standard deviation, derived from the velocity range and the Confidence in Most Likely setting:

Std Dev = (Best Case – Worst Case) × Ratio Scale Multiplier

The Ratio Scale Multiplier is a fixed constant that varies by confidence level: High confidence produces a smaller multiplier (tighter spread); Low confidence produces a larger one (wider spread). If you enter a Standard Deviation Override in Expert Settings, this step is skipped and your value is used directly.

Step 3 — Velocity at selected confidence

Using the mean and standard deviation, the model finds the velocity value that sits at the lower tail of the distribution, corresponding to your selected Confidence in Projected Date:

Velocity at selected confidence = NORM.INV(1 – Confidence, Mean, Std Dev)

At 80% confidence, this finds the velocity exceeded in 80% of scenarios. This is the conservative throughput figure the date calculation is based on.

Step 4 — Sprints and date

With velocity at selected confidence known:

- Sprints needed = Backlog ÷ Velocity at selected confidence
- Business weeks = Sprint Length (weeks) × Sprints needed (rounded)
- Projected Date = Start Date + (Business weeks × 7 days) + Extra Non-Working Days

Sprint rounding uses a deliberate bias toward rounding up. A partial sprint still occupies a full sprint slot in the schedule, so under-rounding would produce dates that are systematically too optimistic.

10. KNOWN LIMITATIONS

Understanding what this model cannot tell you is as important as understanding what it can. These are inherent constraints of the approach, not bugs.

- **Inputs are estimates, not measurements:** The model has no access to your team's actual sprint history. All three velocity values are entered manually. If those inputs are optimistic, the projected dates will be optimistic. The model cannot validate the inputs against reality.
- **No scope change handling:** The model assumes the backlog is fixed from the moment the estimate is set. If requirements are added during the release, the projected date will no longer be accurate. Re-enter an updated backlog to recalculate.
- **Static snapshot:** Each save of a scenario is a point-in-time estimate. The app does not track how the projected date has shifted over time as inputs are updated.
- **Calendar days are approximate:** The projected date is calculated in calendar days (business weeks × 7). Specific weekend dates and regional public holidays are not accounted for automatically — use Extra Non-Working Days to account for known non-working periods.
- **Confidence levels assume a normal distribution:** The model uses a normal (bell-shaped) distribution of velocity outcomes. Real sprint delivery is often more skewed. The asymmetric velocity warning in the app flags cases where this assumption may be strained.
- **Scenarios are independent:** Each scenario is calculated separately. If you have an Optimistic and a Conservative scenario, the app does not blend them or produce a probability-weighted combined estimate. Interpret them as distinct what-if analyses.

— End of Reference Guide —